

Single VO₂ Neuristor Simulations

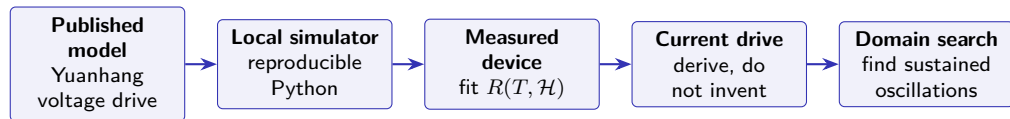
From the Yuanhang voltage-source model to current-source oscillations

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The Central Research Question



Research question

Can the original electrothermal and hysteretic mechanism produce robust relaxation oscillations under an *ideal imposed current*, and where is that operating domain?

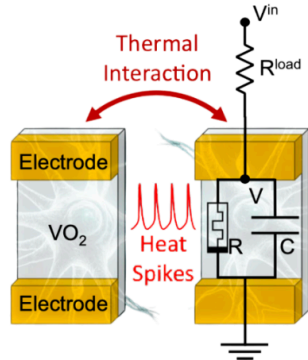
- Preserve the original VO₂ resistance mechanism.
- Reject startup overshoot and timestep-dependent artifacts.
- Visualize internal temperature and resistance dynamics.

Starting Point: Electrical Dynamics

Yuanhang voltage-source node equation

$$C \frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R_{VO_2}}$$

- $C dV/dt$: charge stored at the neuristor node.
- $(V_{in} - V)/R_{series}$: current delivered by the source.
- V/R_{VO_2} : nonlinear leakage through VO_2 .
- $R_{VO_2} = R(T, \mathcal{H})$: temperature and memory couple the electrical and thermal problems.



Voltage-source equivalent circuit.

Starting point: Y.-H. Zhang et al., Nature Communications (2024).

The Feedback Mechanism: Thermal Dynamics

Single-neuristor reduction

$$C_{th} \frac{dT}{dt} = \underbrace{\frac{V^2}{R_{VO_2}}}_{\text{Joule heating}} - \underbrace{S_e(T - T_0)}_{\text{cooling}} + \underbrace{\sigma \eta(t)}_{\text{optional noise}}$$

- C_{th} : thermal inertia of the active device.
- S_e : thermal conductance to substrate/environment.
- Heating lowers R_{VO_2} ; lower resistance changes the voltage and power.
- This delayed negative feedback enables relaxation oscillations.

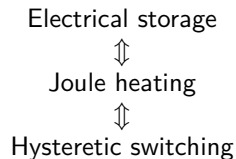
For one isolated device, the inter-neuristor thermal coupling term is zero.

Two coupled state variables

$$V(t), \quad T(t)$$

Material memory

$$\mathcal{H} = \{\delta, T_r, g_r, T_{pr}, \dots\}$$



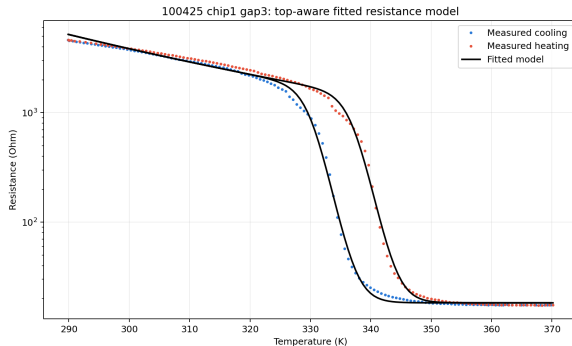
Measured Hysteretic Resistance

$$R(T, \mathcal{H}) = R_0 e^{(E_a/k_B)/T} \underbrace{g(T, \mathcal{H})}_{\text{insulating fraction}} + R_m$$

$$g = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - (T + T_p) \right) \right]$$

- T_c, w, β : center, width, and steepness.
- δ : heating or cooling direction.
- T_p : minor-loop shift after a reversal.
- The same law is used in both source configurations.

Hysteresis: L. A. L. de Almeida et al., Optical Engineering (2002).



Sample 100425 chip1 gap3; log-space $R^2 \approx 0.9987$.

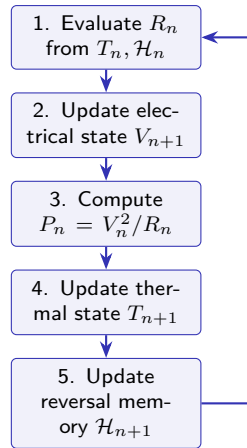
What Happens at Every Timestep?

$$R_n = R(T_n, \mathcal{H}_n)$$

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left(I_{\text{src},n} - \frac{V_n}{R_n} \right)$$

$$T_{n+1} = T_n + \frac{\Delta t}{C_{th}} \left[\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right] + \frac{\sigma}{C_{th}} \sqrt{\Delta t} \zeta_n$$

$$I_{\text{src},n} = \begin{cases} (V_{in,n} - V_n)/R_{series}, & \text{voltage drive,} \\ I_{in,n}, & \text{current drive.} \end{cases}$$



Forward Euler; Euler–Maruyama when $\sigma > 0$.

Production sweeps use $\sigma = 0$.

From Voltage Drive to Ideal Current Drive

Original voltage source

$$C \frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R}$$

I_{src} changes with V

Ideal imposed current

$$I_{in} = C \frac{dV}{dt} + \frac{V}{R}$$

$$C \frac{dV}{dt} = I_{in} - \frac{V}{R}$$

Preserved

Yuanhang $R(T, \mathcal{H})$

single thermal state T

Joule heating and cooling

optional stochastic heating

Assumed in the current-source reduction

one VO₂ device/domain

one thermal node

ideal imposed current

no added phase-lag state

The circuit boundary condition changed; the material mathematics did not.

New Problem: The Oscillatory Domain Was Unknown

Physical controls

I_{in}	imposed electrical drive
C	electrical charging time
C_{th}	thermal inertia
S_e	environmental cooling
T_0	starting/ambient temperature

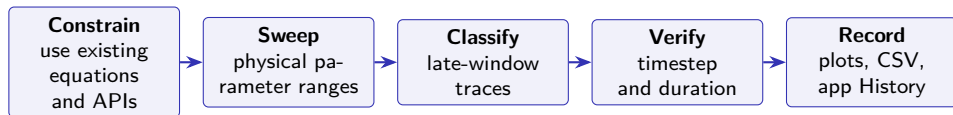
$$\tau_e \sim RC, \quad \tau_{th} \sim \frac{C_{th}}{S_e}$$

Oscillations require enough Joule heating to cross the transition, but enough cooling and electrical recovery to return.

Acceptance criteria

- 1 Oscillations persist in a late-time window.
- 2 More than one full cycle is present.
- 3 Startup overshoot is excluded.
- 4 Frequency is below the numerical/Nyquist limit.
- 5 Period and amplitude survive timestep refinement.
- 6 The path is interpretable on $R(T, \mathcal{H})$.

How I Used Codex as a Research Tool



Representative instruction

“Use the existing current-source simulator without changing the model. Sweep physical values of current, capacitance, thermal capacitance, and thermal conductance. Find sustained oscillations, reject startup overshoot and numerical artifacts, and save successful simulations into app History.”

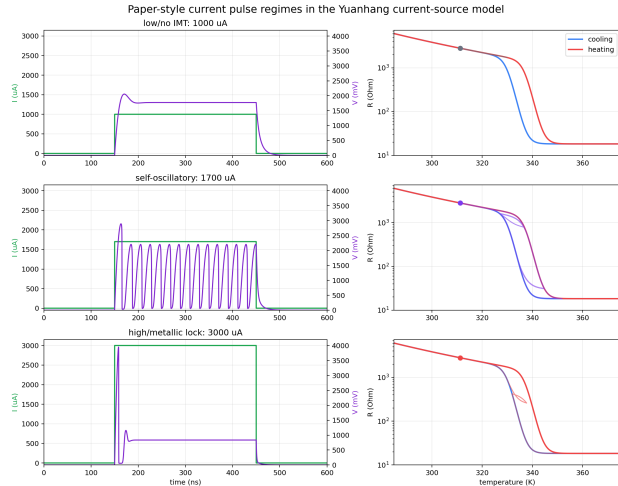
Codex accelerated

- search-script implementation;
- repeated simulations and plots;
- trace metrics and job registration.

Scientific control remained

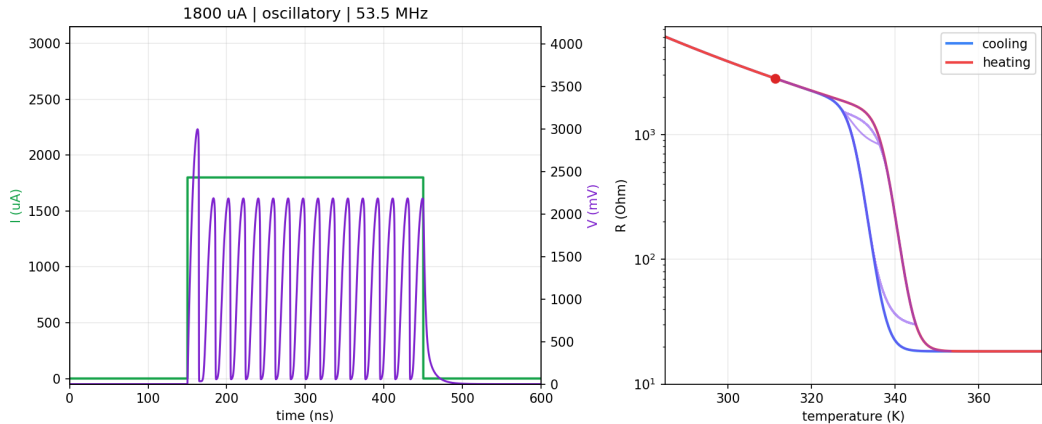
- equations and physical constraints;
- acceptance/rejection criteria;
- interpretation and final claims.

Result: Three Current-Driven Regimes



Low current: insulating equilibrium $\rightarrow$ intermediate current: relaxation oscillation $\rightarrow$ high current: metallic lock.

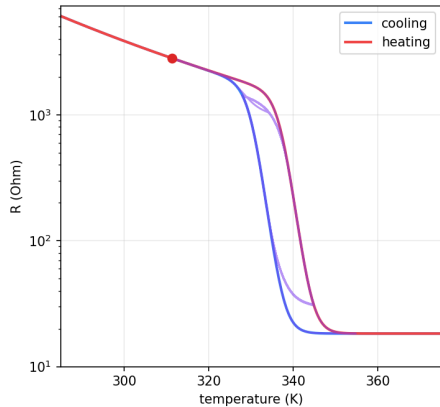
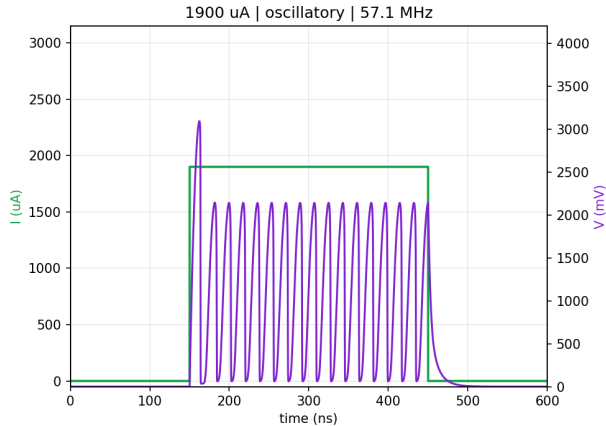
What the Simulation Reveals Inside One Oscillation



1800 μA : sustained 53.5 MHz voltage oscillation. The right panel exposes the otherwise hidden trajectory through the hysteresis loop.

[► Open 10-second time-evolution animation](#)

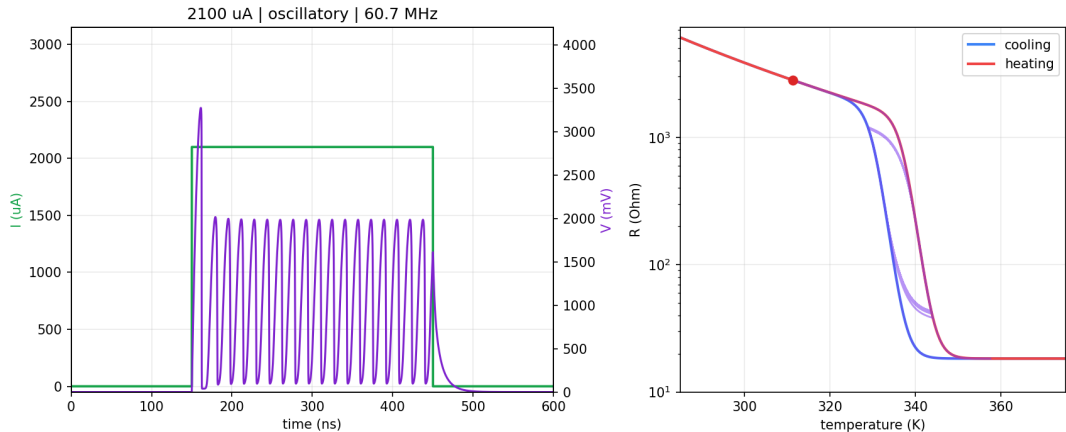
Higher-Frequency Oscillation: 57.1 MHz



1900 μA , $C = 4 \text{ pF}$, $C_{th} = 1 \text{ pJ K}^{-1}$: 57.1 MHz, $V_{pp} \approx 2.08 \text{ V}$, and $E_{\text{cycle}} \approx 39.1 \text{ pJ}$. The trajectory repeatedly closes a stable minor loop.

► [Open high-frequency current-sweep animation](#)

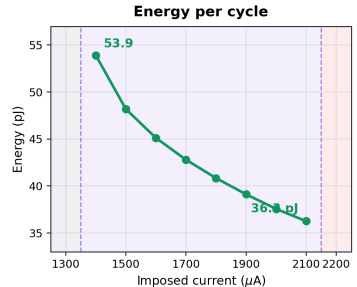
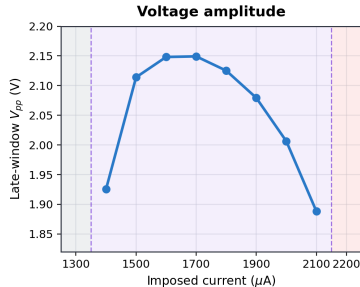
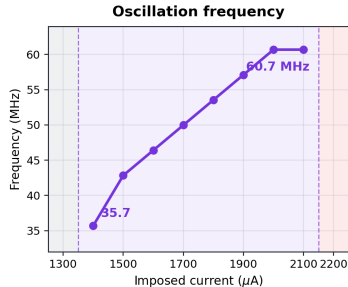
Upper Edge of the Oscillatory Window



At 2100 μA , frequency reaches 60.7 MHz, $V_{pp} \approx 1.89$ V, and $E_{\text{cycle}} \approx 36.3$ pJ. This is the last sustained point before metallic lock at 2200 μA .

Mapped Operating Window

Quantitative response inside the sustained oscillatory domain



Shading: insulating equilibrium | sustained oscillations | metallic lock

Window: 1400 μA to 2100 μA . Frequency increases while cycle energy decreases; voltage amplitude remains approximately 2 V.

$C = 4 \text{ pF}$, $C_{th} = 1 \text{ pJ K}^{-1}$, $S_e = 0.10 \text{ mW K}^{-1}$; current applied from 150 ns to 450 ns.

What Is Established, and What Is Not

Supported by the simulations

- The original Yuanhang hysteresis mechanism can oscillate under ideal current drive.
- Low, oscillatory, and metallic-lock regimes emerge without adding a phase-delay state.
- The paper-frequency analog reaches 35.7 MHz to 60.7 MHz.
- Serial and vectorized traces agree; representative periods converge with smaller Δt .

Not yet an absolute experimental fit

- The simulated current window is higher than the measured one.
- Cycle energy is ~ 36 pJ to 54 pJ, not the paper's ~ 1 pJ to 2 pJ.
- C , C_{th} , S_e , and source non-ideality need transient calibration.
- The model is phenomenologically corroborative, not parameter-identical.

Conclusions and Next Steps

- 1 **Calibrate dynamics:** fit C , C_{th} , S_e , and a non-ideal current-source output impedance to transient data.
- 2 **Validate across samples:** compare onset current, voltage floor, frequency, amplitude, and cycle energy.
- 3 **Quantify uncertainty:** report parameter intervals and sensitivity, not only best-fit traces.
- 4 **Preserve reproducibility:** keep equations hardcoded, regression-tested, and every accepted run in public app History.

Takeaway

We rebuilt and validated the published VO_2 neuristor model, derived its ideal-current counterpart, and used AI-assisted but physically constrained domain search to demonstrate robust current-driven relaxation oscillations.

Thank you

Backup: Representative Timestep Convergence

Δt (ns)	Period (ns)	Frequency (MHz)
0.015820	123.206	8.116
0.007910	123.135	8.121
0.003955	123.095	8.124

Interpretation

The period approaches a stable value as the timestep is halved. This is the expected signature of a physical solution being resolved more accurately, rather than an oscillation created by the numerical grid.

Representative Yuanhang current-source case: $I = 1640.226 \mu\text{A}$.

Backup: Final Current-Source Standard

$$\begin{aligned} C \frac{dV}{dt} &= I_{in} - \frac{V}{R}, \\ C_{th} \frac{dT}{dt} &= \frac{V^2}{R} - S_e(T - T_0), \\ R(T, \mathcal{H}) &= R_m + R_0 e^{(E_a/k_B)/T} g(T, \mathcal{H}). \end{aligned}$$

- Exactly one VO₂ device/domain and one thermal node.
- Upstream-faithful Yuanhang/Almeida hysteresis and reversal memory.
- Deterministic default $\sigma = 0$; optional stochastic heating remains available.
- Obsolete delayed-phase, double-thermal, and multidomain variants are not active models.

Backup: Reproducibility Trail

- Fitted sample: 100425 chip1 gap3.
- Public History job: 20260707_145140_paper_frequency_f881d2.
- Public History job: 20260707_145140_sample_scale_89ffb4.
- Each job stores parameters, trace CSVs, spectral summaries, figures, and the animation.
- Current-source regression tests cover finite traces, serial/vectorized equality, old-history compatibility, and voltage-source preservation.

AI-use principle

Codex was allowed to automate implementation and search, but successful results were accepted only after model-preservation, physical interpretation, and numerical-convergence checks.